Density Functional Theory: General Concepts

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Outline

- Why computational materials science?
- Time and length scales
- Why ab initio/DFT calculations?
- The electron density as the fundamental quantity
- Formalism, theorems, and the Kohn-Sham equations
- Approximations for the exchange-correlation functional
- Pseudopotentials
- Plane-wave expansion, Brillouin zone sampling
- Basic properties with DFT

HPC for materials science



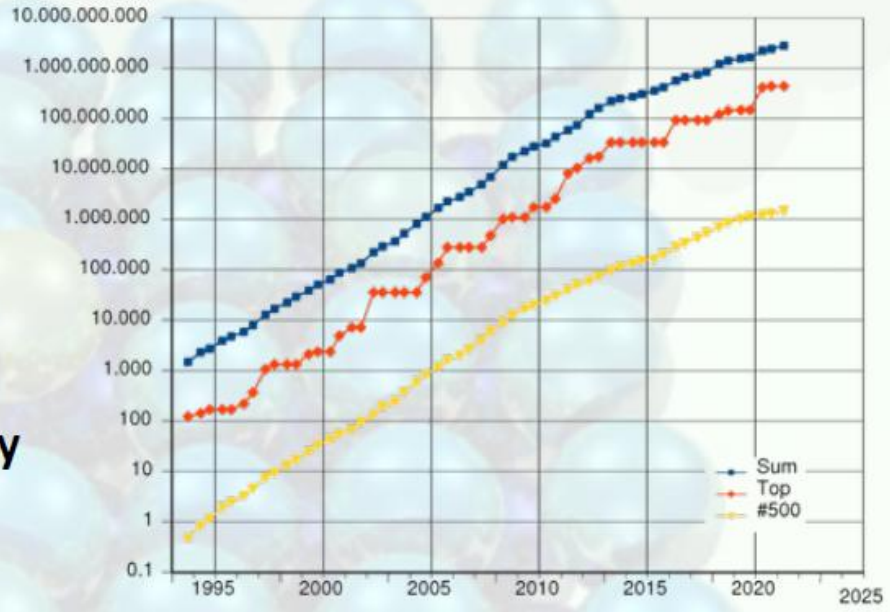


HPC for materials science

THE RISE OF COMPUTATIONAL SCIENCE

A calculation that took **one year in 1993** takes **30 seconds in 2022** (1-million-fold increase).

And this is just with bits: neurons are in, and qubits on the horizon. **21st-century science and discovery will be driven by computational science.**

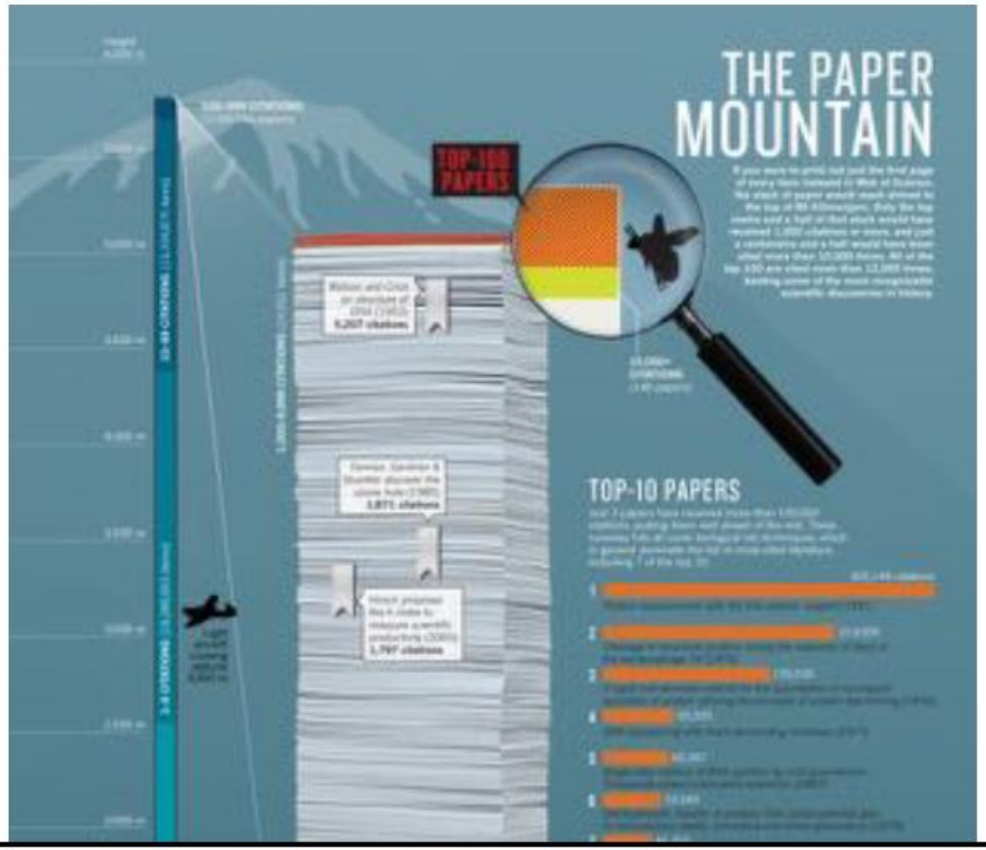


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IMPACT OF COMPUTATIONAL QUANTUM MECHANICS



THE TOP 100 PAPERS:

12 papers on density-functional theory in the top-100 most cited papers in the entire scientific-medical-engineering literature, ever.

NATURE, OCT 2014

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MATERIALS ARE KEY TO SOCIETAL WELL BEING

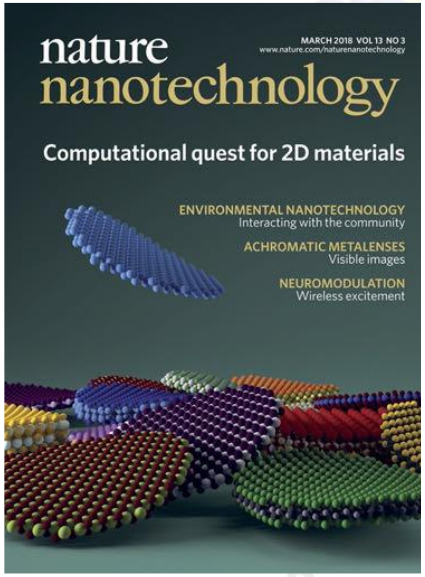
We need novel materials for:

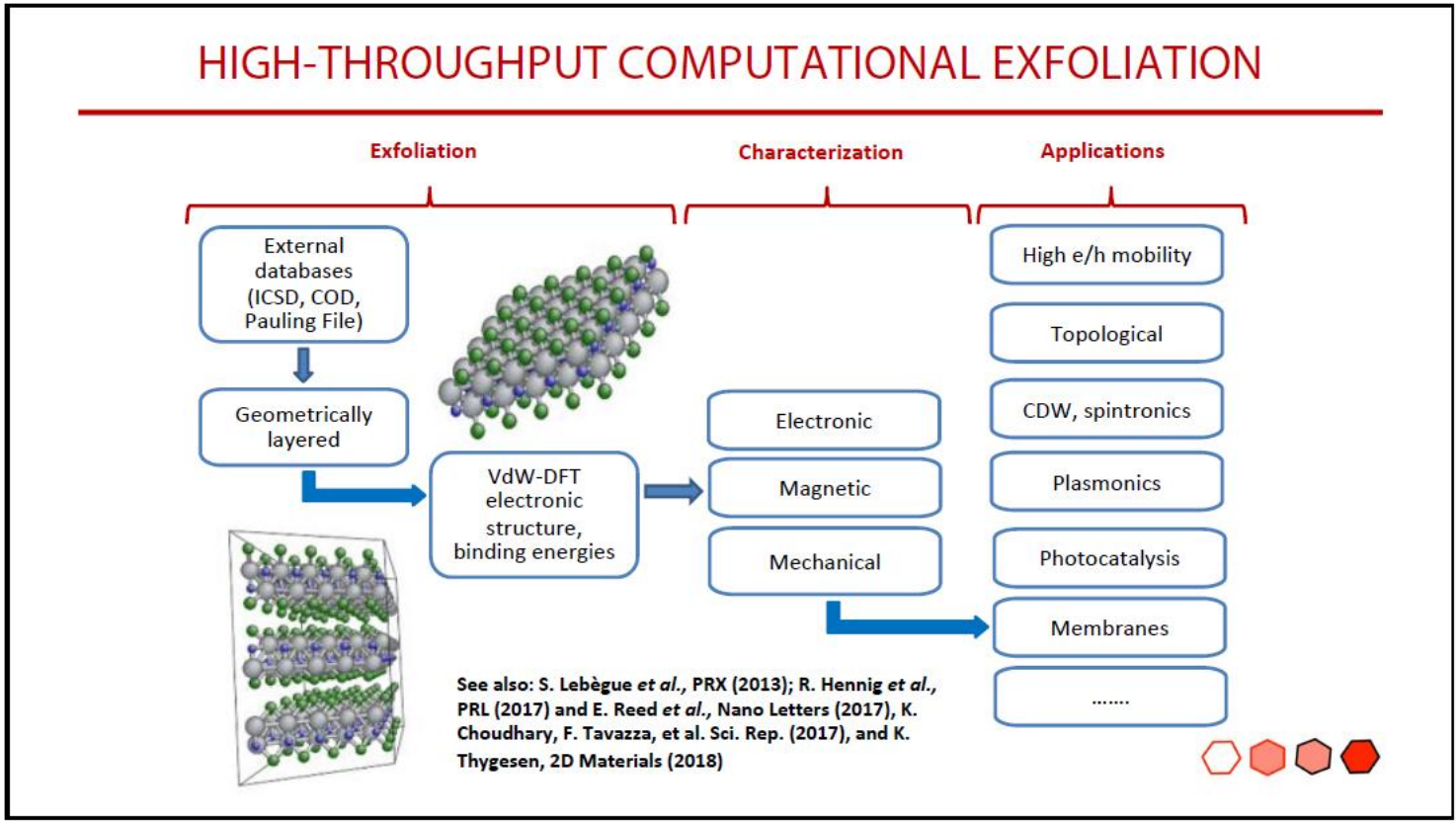
- **Energy harvesting, conversion, storage, efficiency**
- **Environmental protection and reparation**
- **High-tech and high-value industries**
- **Information and communication technologies**
- **Health care and biomedical engineering**
- **Pharmaceuticals** (crystallization, stability, polytypes)
- **Monitoring, provenance, and safety of foods**
- **Fundamental science** (graphene and 2D materials, topological insulators, entangled spins for quantum computing, high-T_c)
- **Experimental science** (detectors, sensors, magnets)



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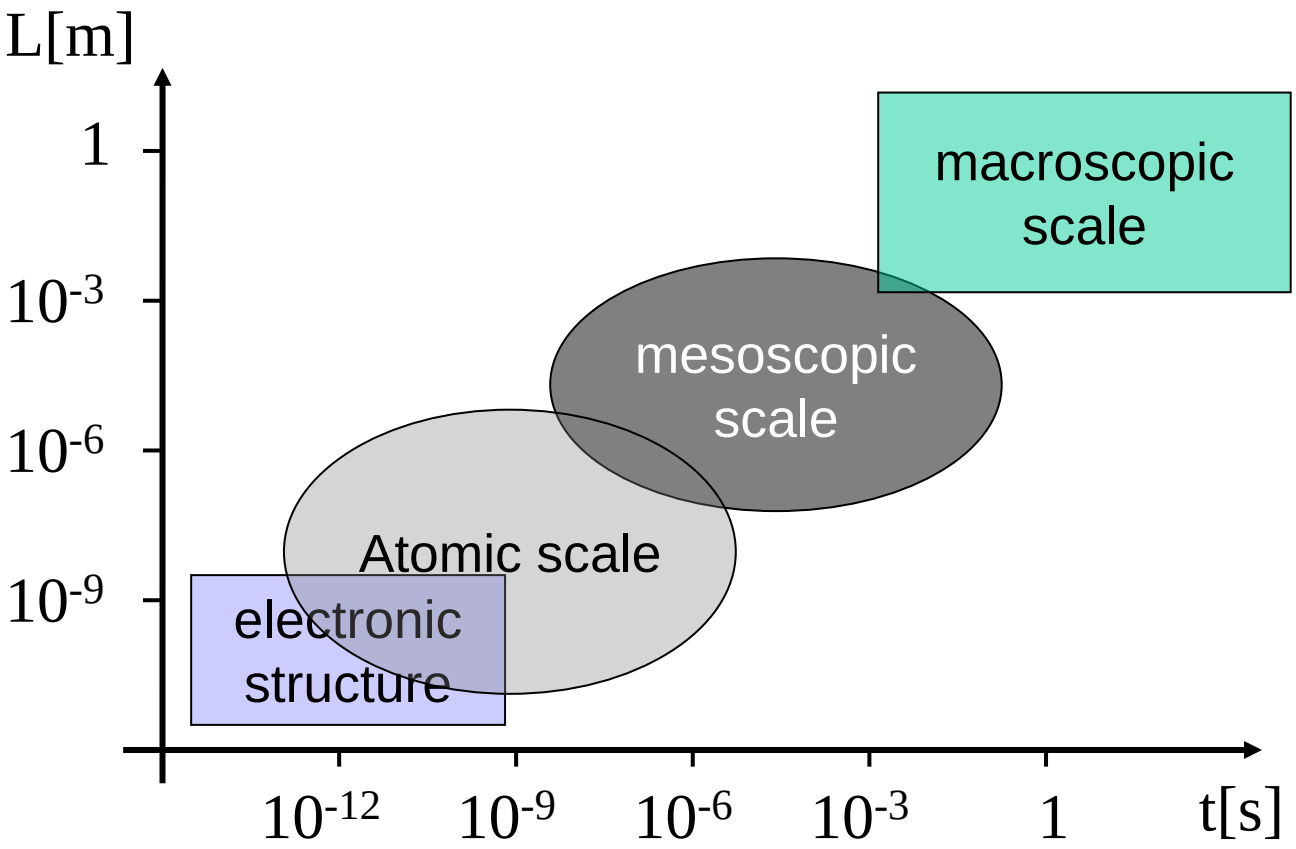
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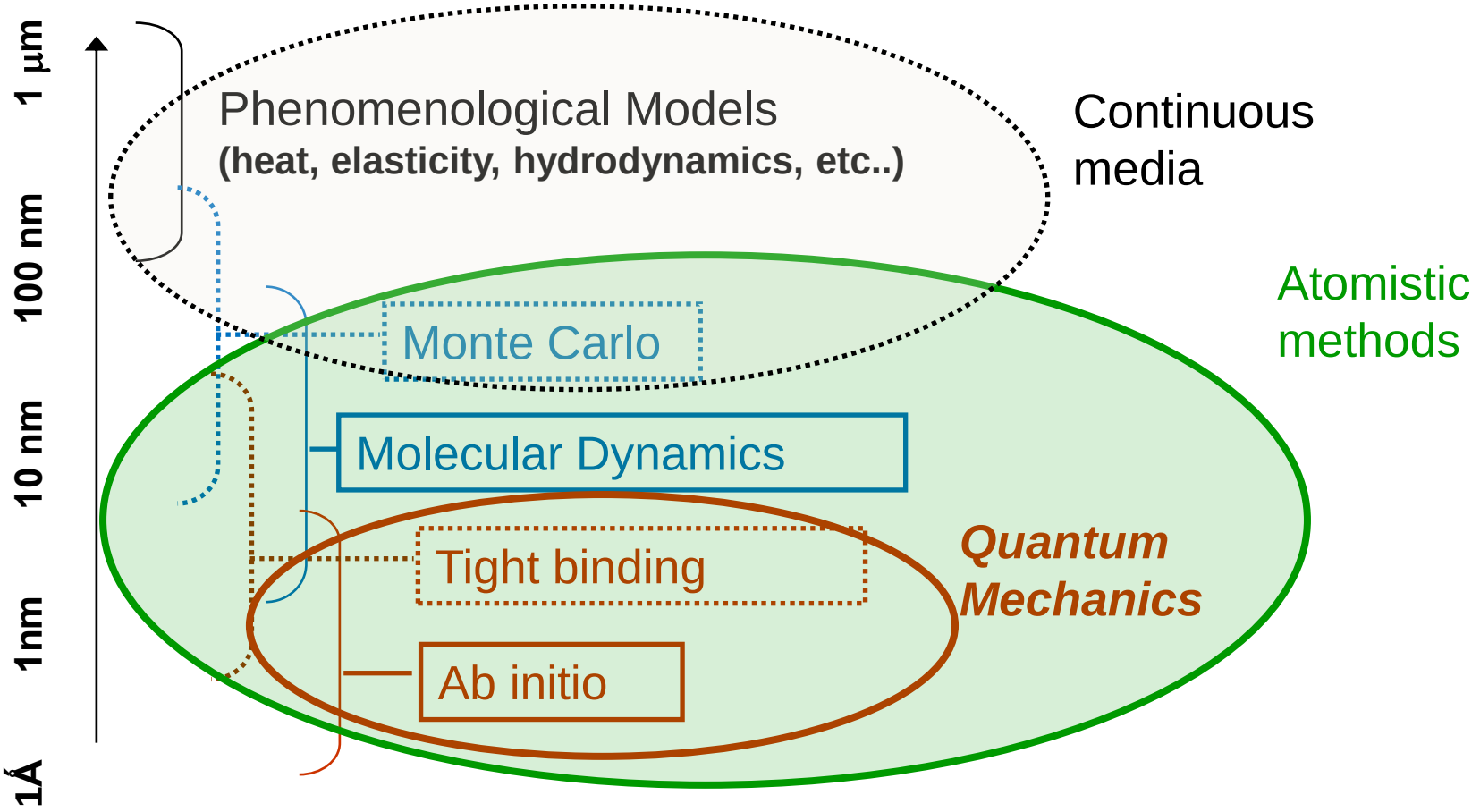


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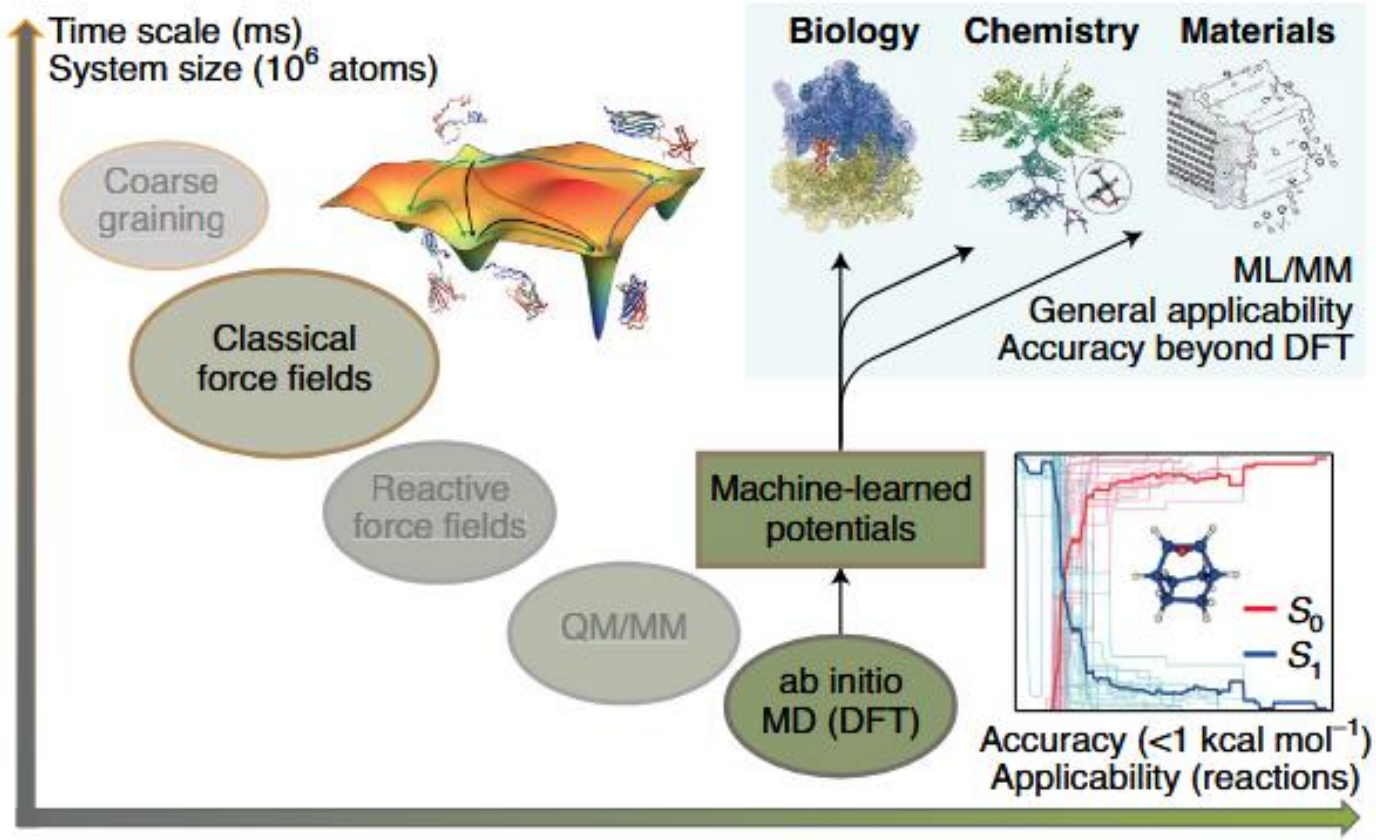
Length/time scales, methods



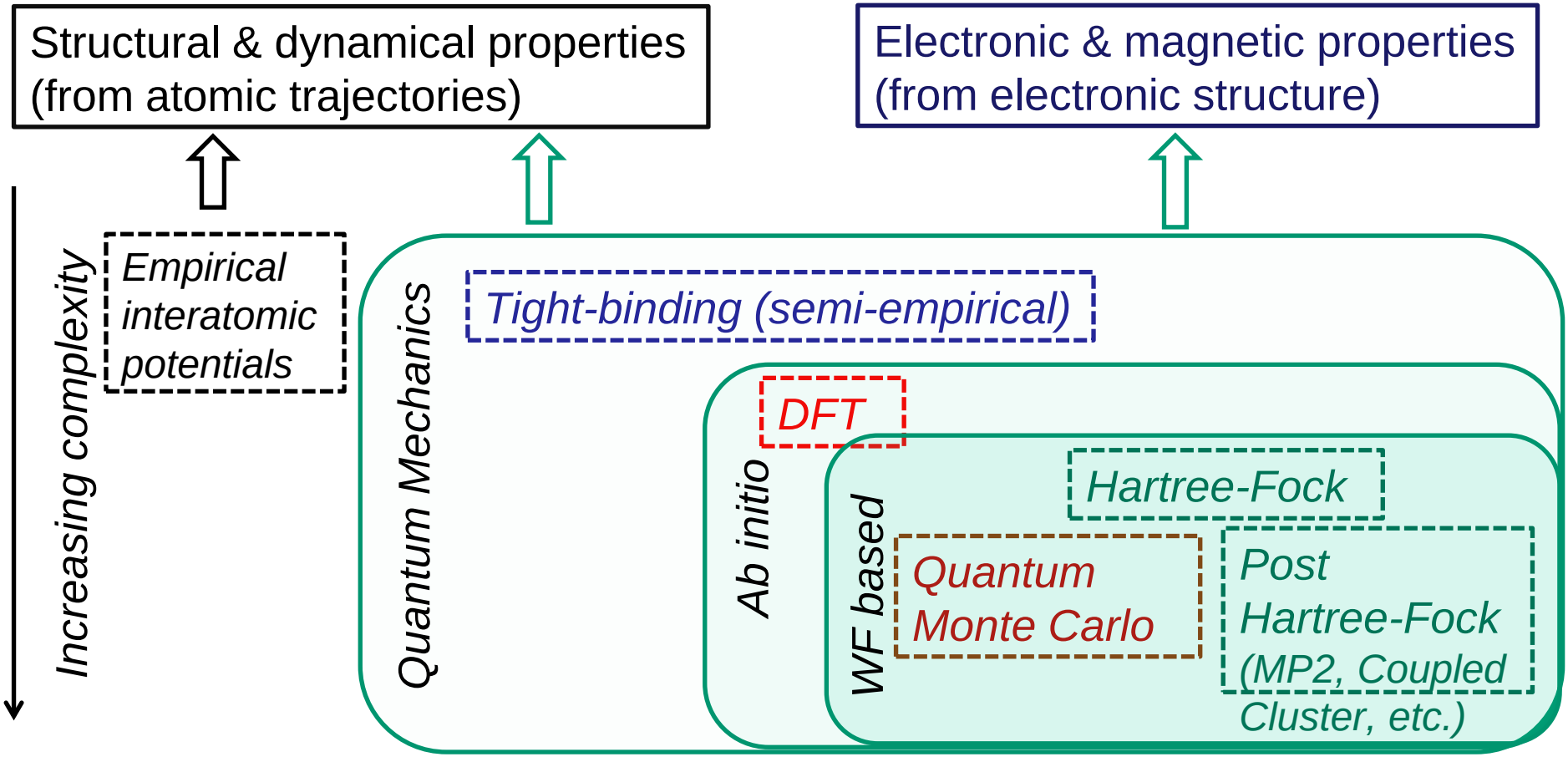
Length/time scales, methods



Length/time scales, methods



Length/time scales, methods



Classical methods: major drawbacks

- Potential tailored on specific properties of the system
- Potential parameters taken from experiments
- Classical potentials not transferable
- Chemical reactions, change of bonding properties and in general electronic properties not described
- Ab initio calculations: treating explicitly the electronic degrees of freedom

Ab initio vs classical calculations

- Ab initio interactions solely determined by the electronic structure of atoms
- Solution of the quantum mechanical EOM of the electrons: energy and forces
- Atomic number the only “external” input: transferability
- However, a special theoretical framework is needed

Ab initio calculations: why?

- Same approach from biology to planet interiors
- Electronic structure and properties, chemical reactions, change of coordination
- Accuracy at the (sub)nano scale
- Directly simulate experimental spectra (x-ray, Raman, infrared, NMR,…)

Ab initio calculations: why?

Generality (« transferability »): all atoms from periodic table **Desired accuracy** (« predictive » theory):

- cristallographic positions ± 0.005 Å
- bonding energies ± 40 meV/atom (≅ 1 kcal/mol)
- vibrational frequences 100 GHz ~ 3.3 cm^{-1}
- dipolar moments ± 0.02 D
- magnetic moments ± 0.1 μ_B
- spectra (visible, UV, X) ± 0.1 eV

Typical system size: < 10^3 atomes

Typical time scale: ~ 10^{-10} s

Computational efficiency: parallel algorithms, modularity

The many-electron problem

Time-independent, non-relativistic, spinless, Schrödinger equation for N electrons interacting with fixed nuclei

$$H\Psi(\vec{r}_1,\ldots,\vec{r}_N) = \varepsilon\Psi(\vec{r}_1,\ldots,\vec{r}_N)$$

$$H = \sum_{i=1}^{N}\left[-\frac{\hbar^2\nabla_i^2}{2m} + V_{\text{ext}}(\vec{r}_i)\right] + \sum_{i>j}\frac{e^2}{|\vec{r}_i - \vec{r}_j|}$$

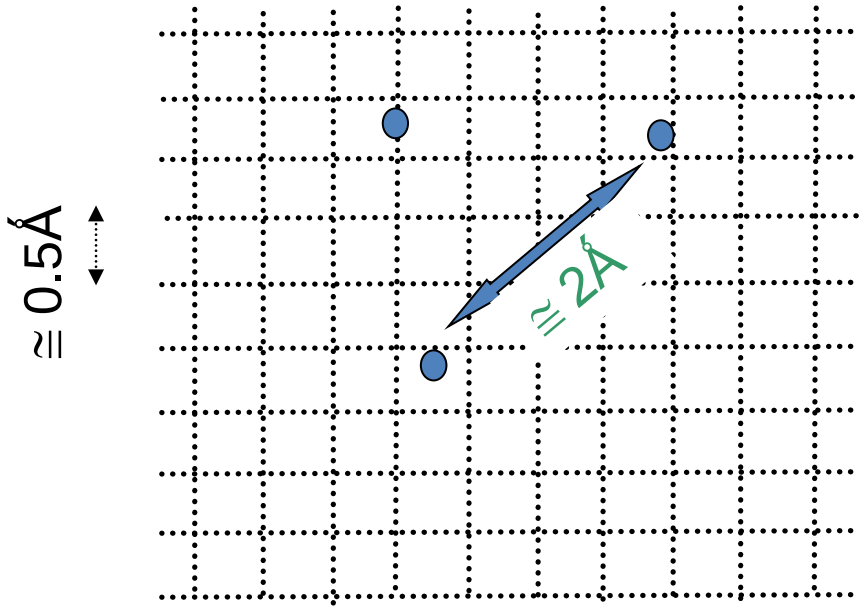
sum of *N* one-electron operators (kinetic operator + Coulomb potential from the nuclei)

Long range e^- – e^- interaction (the big problem!)

Can we solve it numerically by brute force?

The many-electron problem by brute force?

Example: Li_3 cluster



9 electrons

Solve the Schrödinger equation on a MxMxM grid <u>for each spatial coordinate</u>

For M=10 → 10^{27} x 2^9 independent variables

→ ~ 5 x 10^{29} operations just to evaluate $\langle \Psi | H | \Psi \rangle$!

On a Tflops computer this would need ~ 10^{11} years!
(unfeasible even if computer speed increases by a factor ~4 every ~2 yrs)

Adiabatic (Born-Oppenheimer) approximation

- Electrons much lighter & faster than nuclei, their timescales much shorter
- Quantum EOM for electrons, classical ones for ions
- Solution of the electrons QM: global many-body wavefunction, whose size grows exponentially

The electronic density

$$n(\vec{r}) = \langle\Psi|\sum_{i=1}^{N}\delta(\vec{r}-\vec{r}_i)|\Psi\rangle = N\int d^3r_2\ldots\int d^3r_N\,|\Psi(\vec{r},\vec{r}_2,\ldots,\vec{r}_N)|^2$$

- Is a function of 3 spatial coordinates, independently of *N*
- Is measurable
- Many physical quantities can be inferred from it

Can we reformulate the many-body electron problem in terms of *n(r)*?

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The Hohenberg-Kohn theorem (aka Density Functional Theory)

PHYSICAL REVIEW VOLUME 136, NUMBER 3B 9 NOVEMEBR 1964

Inhomogeneous Electron Gas*

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(Received 18 June 1964)

The Hohenberg-Kohn theorem (aka Density Functional Theory)

1) For non-degenerate ground states: one-to-one correspondence between Hamiltonians *H* and ground-state electron densities $n_0(r)$

the ground-state energy is a functional of $n(r)$: $E = E[n]$

2) We consider a large class of "physically sensible" electron densities $n(r)$; $E[n]$ is minimal when $n(r)$ is the actual ground-state density $n_0(r)$:

$\min E[n] = E[n_0]$

Can we thus "just" minimize $E[n]$?

The energy functional

$$H = \sum_{i=1}^{N} \left[-\frac{\hbar^2 \nabla_i^2}{2m} + V_{\text{ext}}(\vec{r}_i) \right] + \sum_{i>j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|}$$

$$E[n] = T[n] + E_{\text{ext}}[n] + U[n]$$

$$E_{\text{ext}}[n] = \int d^3r \, V_{\text{ext}}(\vec{r}) \, n(\vec{r})$$

$$U[n] = E_{\text{H}}[n] + \Delta U[n] \qquad E_{\text{H}}[n] = \frac{e^2}{2} \int d^3r \int d^3r' \, \frac{n(\vec{r}) n(\vec{r'})}{|\vec{r} - \vec{r'}|}$$

Kohn-Sham: map onto a non-interacting system

- **Separate, in *T[n]*, the kinetic energy of *N* non-interacting electrons but not necessarily homogeneous $T_s[n]$**

$$T[n] = T_s[n] + T_c[n]$$

- **Add the term E_{XC} that describe quantum effects (exchange and correlation) and the difference between *T[n]* and $T_s[n]$**

$$E_{xc}[n] = T_c[n] + \Delta U[n]$$

The energy functional rewritten

$$E[n] = T[n] + E_{\text{ext}}[n] + U[n]$$

$$U[n] = E_{\text{H}}[n] + \Delta U[n]$$

$$T[n] = T_s[n] + T_c[n]$$

$$E_{\text{xc}}[n] = T_c[n] + \Delta U[n]$$

$$E_{\text{ext}}[n] = \int d^3r \, V_{\text{ext}}(\vec{r}) \, n(\vec{r})$$

$$E_{\text{H}}[n] = \frac{e^2}{2} \int d^3r \int d^3r' \, \frac{n(\vec{r}) n(\vec{r'})}{|\vec{r} - \vec{r'}|}$$

$$E[n] = T_s[n] + \int d^3r \, V_{\text{ext}}(\vec{r}) \, n(\vec{r}) + E_{\text{H}}[n] + E_{\text{xc}}[n]$$

The Kohn-Sham equations

- **Regroup all "potential" terms into a single-particle effective potential**

$$V_s(\vec{r}) = V_{ext}(\vec{r}) + e^2 \int d^3r' \, \frac{n(\vec{r'})}{|\vec{r} - \vec{r'}|} + V_{xc}(\vec{r}; [n])$$

$$V_{xc}(\vec{r}; [n]) = \delta E_{xc}[n] / \delta n(\vec{r})$$

- **Rewrite Schrödinger equation in terms of (solvable) independent single-particle equations**

$$\left[-\frac{\hbar^2 \nabla^2}{2m} + V_s(\vec{r}) \right] \psi_j(\vec{r}) = \varepsilon_j \psi_j(\vec{r}) \qquad n_s(\vec{r}) = \sum_j f_j \; | \psi_j(\vec{r}) |^2$$

The exchange-correlation term

- **Contains everything we do not know**

$$T[n] = T_s[n] + T_c[n] \qquad U[n] = E_{\mathrm{H}}[n] + \Delta U[n]$$

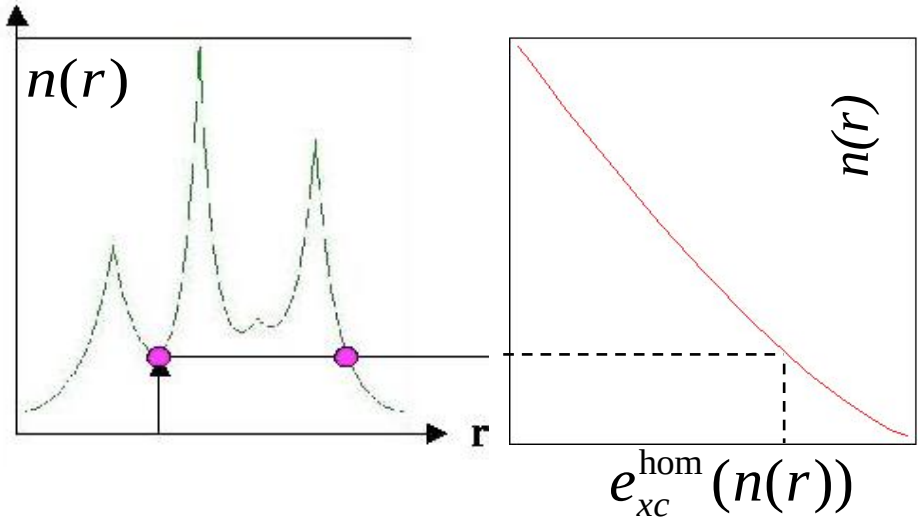
$$E_{\mathrm{xc}}[n] = T_c[n] + \Delta U[n]$$

  - **The kinetic energy difference between the interacting and non-interacting systems at the same density *n(r)***
  - **Quantum effects on e-e interactions: exchange, Coulomb correlation, self-interaction correction**

DFT and the local density approximation (LDA)

- Unknown electron-electron interaction confined in the exchange-correlation term
- DFT is an exact and useless ground state theory, no analytical form of this $E_{XC}[n]$ functional
- LDA: the XC term described as a function $E_{XC}(n)$ of the density
- Its numerical expression through exact quantum Monte Carlo simulations of the homogenous electron gas

DFT and the local density approximation (LDA)



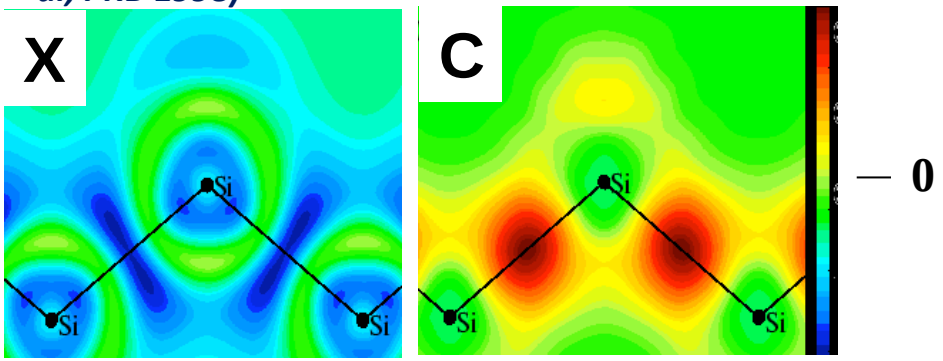
- The inhomogeneous system is considered as locally homogeneous
- The XC energy per particle is the same as in the homogeneous electron gas

Ceperley & Alder, PRL 1980

DFT and the local density approximation (LDA)

- Despite its simplicity, LDA works surprisingly well
- Sum rule: the XC hole contains exactly one electron
- Error compensation: LDA is exact in the limit N/V –> ∞

Difference between the exact XC energy and LDA energy in c-Si *(Hood et al, PRB 1998)*



LDA drawbacks

- Electrons not enough localized (not too surprising, because the LDA is exact only for completely delocalized electrons in the HEG)
- Strongly correlated electron systems are very badly described
- Self-interaction problem
- Over-binding in most cases (bonding energy is often overestimated, bond distance underestimated)
- “Weak” interactions: H-bonds, van der Waals

E_{XC}: the Generalized Gradient Approximation (GGA)

- GGA: improving LDA by including the derivatives (gradients) in the XC functional

$$E_{\mathrm{xc}}^{\mathrm{GGA}}[n,\sigma] = E_{\mathrm{xc}}^{\mathrm{LDA}}[n,\sigma] + \int d^3r\, e_{\mathrm{xc}}^{\mathrm{GGA}}\left(n(\vec{r}), \nabla n(\vec{r})\right)$$

- Many available recipes:
  - Becke (exchange) – Perdew (correlation): BP86
  - Becke (X) – Lee-Yang-Parr (correlation): BLYP
  - Perdew-Burke-Ernzerhof: PBE

E_{XC}: the Generalized Gradient Approximation (GGA)

- GGAs tend to overcorrects LDA: bond lengths ~ 0-2% larger than experimental values, cohesive energy 10-20% too small
- GGAs cannot describe long-range (wan der Waals) effects; H-bonds are usually well accounted for
- GGAs describe XC effects in small-density regions of space much better than LDA; usually employed for atoms, molecules, surfaces
- Strongly correlated electron systems are still out of reach

E_{XC}: the Jacob's ladder of XC functionals

E_{XC} depends on:

Hybrid Functionals	$n(r)\,,\nabla n(r)\,,\{\varphi_i(r)\}$
Meta-GGA	$n(r)\,,\nabla n(r)\,,\{\nabla^2\varphi_i(r)\}$
DFT-vdW including dispersion via the response function to perturbations	$n(r)\,,\nabla n(r)\,,\frac{\delta n(r)}{\delta V(r')}$
Generalized Gradient Approximations (GGA)	$n(r)\,,\nabla n(r)$
Local Density Approximations (LDA, LSDA)	$n(r)$

DFT and excitations

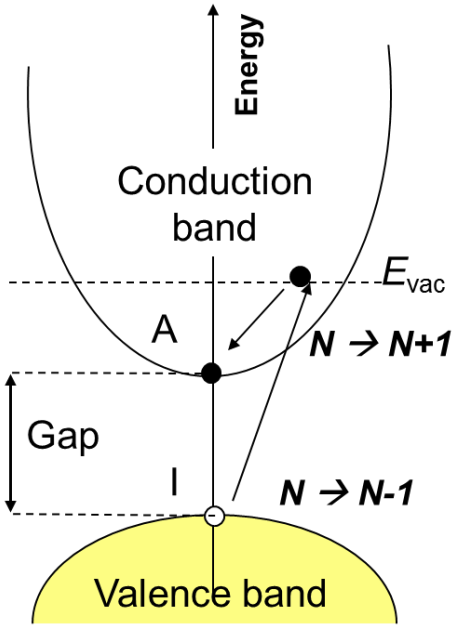
- DFT (no matter which XC functional is used) is a ground-state theory
- Kohn-Sham eigenvalues of the independent-particle equations are NOT the interacting-electrons values

$$\left[-\frac{\hbar^2\nabla^2}{2m}+V_{\rm s}(\vec{r})\right]\psi_j(\vec{r})=\varepsilon_j\psi_j(\vec{r})$$

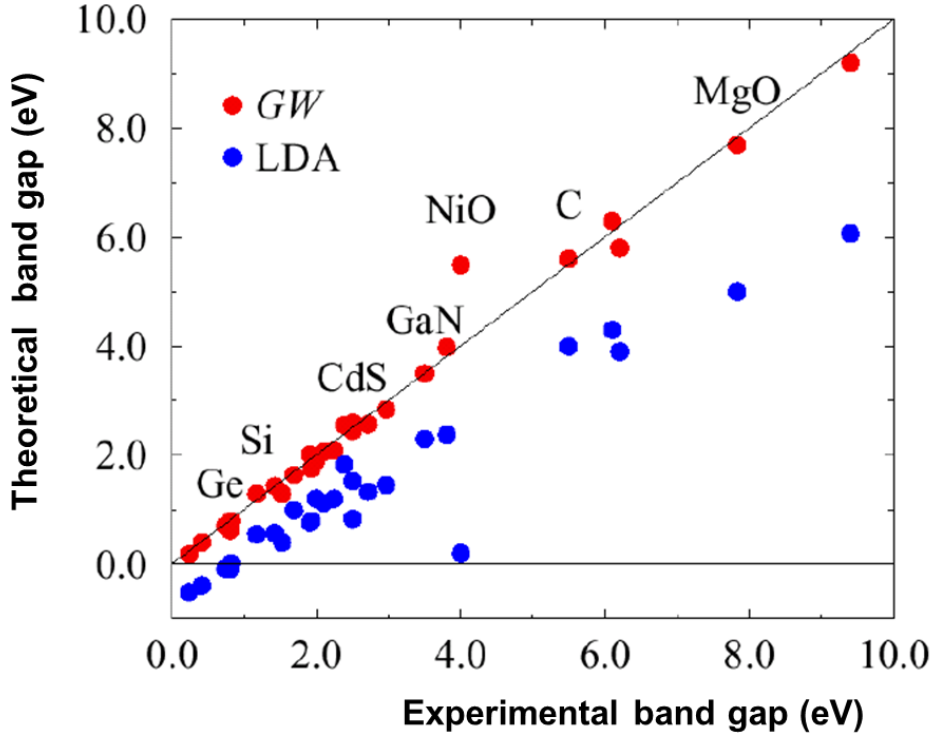
- They usually reproduce quite well the valence electronic bands
- They are badly, badly wrong in describing excited states, for example the conduction bands

DFT and excitations

The band gap problem: most approximations to the exact functional systematically underestimate band gaps

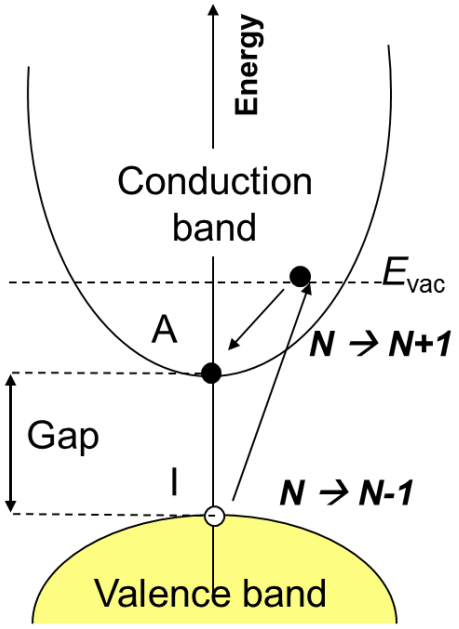


Gap = I - A



DFT and excitations

The band gap problem: most approximations to the exact functional systematically underestimate band gaps



Gap = I - A

$$E_{gap} = I - A = (E_{vac} - \varepsilon_N^{(N)}) - (E_{vac} - \varepsilon_{N+1}^{(N+1)}) =$$

$$= \varepsilon_{N+1}^{(N+1)} - \varepsilon_N^{(N)} = \varepsilon_{N+1}^{(N+1)} - \varepsilon_{N+1}^{(N)} + \varepsilon_{N+1}^{(N)} - \varepsilon_N^{(N)}$$

This is the variation of *(N+1)th* eigenvalue when changing the number of electrons. Cannot be obtained straightforwardly within a theory that is devised for fixed *N*

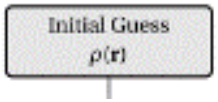
This is the « DFT » gap: LUMO-HOMO at constant number *N* of electrons

DFT: synthesis and take-home messages

- An effective theory for ground-state properties, through the minimization of the energy, which is a functional of the electron density
- A way to recast the interacting many-electron problem into a set of self-consistent one-electron equations
- In principle based only on the electron density; actually, single-particle orbitals are used in the determination of the electron density
- Exact in principle, but necessarily approximated (the exact $E_{XC}[n]$ is unknown)
- Unfit to study excited-state properties; usually a starting point for more sophisticated approaches

DFT: practical implementation

A. Initialization



- **Set up the basic parameters (unit cell, atomic positions, k points, cutoff energy, atom types with the corresponding pseudo-potentials, XC approximation, etc.)**
- **Choose a starting electron density (ex: superposition of atomic-like densities)**

B. Self-consistent cycle

Numerically solve the Kohn-Sham equations

- **Kohn-Sham equations**

$$h^{\mathrm{KS}}[n]\,\psi_j(\vec{r}) = \varepsilon_j\,\psi_j(\vec{r}) \qquad h^{\mathrm{KS}}[n] = -\frac{\hbar^2\nabla^2}{2m} + V_{\mathrm{ext}}(\vec{r}) + e^2\int d^3r'\,\frac{n(\vec{r'})}{|\vec{r}-\vec{r'}|} + V_{\mathrm{xc}}(\vec{r};[n])$$

$$n(\vec{r}) = \sum_j f_j\ |\,\psi_j(\vec{r})\,|^2$$

- **Wavefunctions are expanded on a basis**

$$\psi_j(\vec{r}) = \sum_\alpha c_{j,\alpha} f_\alpha(\vec{r}) \qquad h^{\mathrm{KS}}[n]\sum_\alpha c_{j,\alpha} f_\alpha(\vec{r}) = \varepsilon_j \sum_\alpha c_{j,\alpha} f_\alpha(\vec{r})$$

- **The problem becomes a linear algebra one**

$$\sum_\alpha c_{j,\alpha}\int d^3r\ f^*_\beta(\vec{r})\,h^{\mathrm{KS}}[n]\,f_\alpha(\vec{r}) = \sum_\alpha c_{j,\alpha}\varepsilon_j \int d^3r\ f^*_\beta(\vec{r})\,f_\alpha(\vec{r})$$

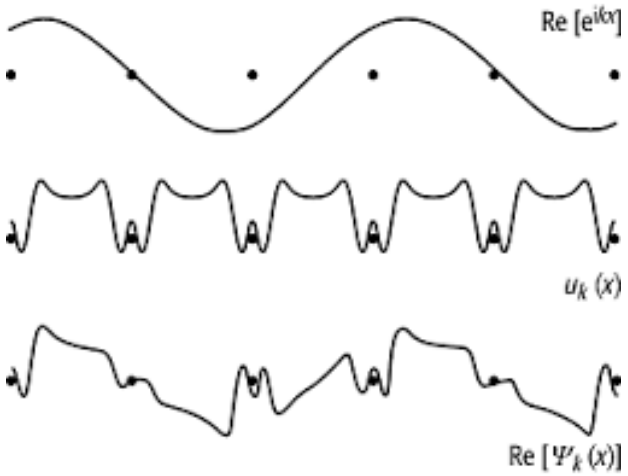
$$\sum_\alpha c_{j,\alpha}\left[h^{\mathrm{KS}}_{\alpha,\beta} - \varepsilon_j\,S_{\alpha,\beta}\right] = 0 \qquad \mathrm{Det}\left[\mathcal{H}^{\mathrm{KS}} - \varepsilon_j\,\mathcal{S}\right] = 0$$

Periodic crystals/cells: plane-wave expansion

- **Wave-functions must have the same periodicity of the crystal (Bloch's theorem)**

$$\psi_{m,\vec{k}}(\vec{r}+\vec{r}_l) = \mathrm{e}^{\mathrm{i}\,\vec{k}\cdot\vec{r}_l}\,\psi_{m,\vec{k}}(\vec{r})$$

$$\psi_{m,\vec{k}}(\vec{r}) = \frac{1}{\sqrt{N\Omega_0}}\,\mathrm{e}^{\mathrm{i}\,\vec{k}\cdot\vec{r}}\,u_{m,\vec{k}}(\vec{r})$$



Periodic crystals/cells: plane-wave expansion

- Wave-functions must have the same periodicity of the crystal (Bloch's theorem)

$$\psi_{m,\vec{k}}(\vec{r}+\vec{r}_l) = \mathrm{e}^{\mathrm{i}\,\vec{k}\cdot\vec{r}_l}\,\psi_{m,\vec{k}}(\vec{r}) \qquad \psi_{m,\vec{k}}(\vec{r}) = \frac{1}{\sqrt{N\Omega_0}}\,\mathrm{e}^{\mathrm{i}\,\vec{k}\cdot\vec{r}}\,u_{m,\vec{k}}(\vec{r})$$

- They can be expanded over the reciprocal lattice

$$u_{m,\vec{k}}(\vec{r}) = \sum_{\{\vec{G}\}} \mathrm{e}^{\mathrm{i}\,\vec{G}\cdot\vec{r}}\,\tilde{u}_m(\vec{k}+\vec{G})$$

- Plane-wave form an infinite set: a cutoff must be chosen

$$u_{m,\vec{k}}(\vec{r}) = \sum_{\{\vec{G}\}:\frac{\hbar^2|\vec{k}+\vec{G}|^2}{2m}\leq E_{\mathrm{cut}}} \mathrm{e}^{\mathrm{i}\,\vec{G}\cdot\vec{r}}\,\tilde{u}_m(\vec{k}+\vec{G})$$

Numerically solve the Kohn-Sham equations

- KS wave-functions are expressed in terms of plane-waves

$$\psi_{m,\vec{k}}(\vec{r}) = \sum_{\{\vec{G}\}} \mathrm{e}^{\mathrm{i}\,(\vec{k}+\vec{G})\cdot\vec{r}}\ \tilde{u}_m(\vec{k}+\vec{G})$$

- The electron density reads

$$n(\vec{r}) = 2\sum_{m}\sum_{\vec{k}} w_{\vec{k}}\ f_{m,\vec{k}}\ \mid \psi_{m,\vec{k}}(\vec{r})\ \mid^2$$

- Integrals in the *k*-space are replaced by a sum of finite special symmetry points (see later)

$$\bar{f} = \frac{\Omega}{(2\pi)^3}\int_{\mathrm{BZ}} d^3k\ f(\vec{k}) \ \simeq\ \sum_{i=1}^{N_{\vec{k}}} w_{\vec{k}_i} f(\vec{k}_i)$$

Numerical solutions: making us of direct and reciprocal spaces and Fast Fourier Transforms

$$n(\boldsymbol{r}) = 2\,\Sigma_{m,\boldsymbol{k}}\; f_m\, w_{\boldsymbol{k}}\, |\psi_{m\boldsymbol{k}}(\boldsymbol{r})|^2 \qquad \psi_{m\boldsymbol{k}}(\boldsymbol{r}) \sim \Sigma_{\boldsymbol{G}}\, e^{\,i\,(\boldsymbol{k}+\boldsymbol{G})\,\boldsymbol{r}}\; u_{m,\,\boldsymbol{k}+\boldsymbol{G}}$$

Naively: $n(\boldsymbol{r}) \sim 2\,\Sigma_{m,\boldsymbol{k}} f_m w_{\boldsymbol{k}}\;\; \Sigma_{\boldsymbol{G},\boldsymbol{G}'}\, e^{i\,(\boldsymbol{G}-\boldsymbol{G}')\,\boldsymbol{r}}\, u_{m,\boldsymbol{k}+\boldsymbol{G}}\, u^*_{m,\,\boldsymbol{k}-\boldsymbol{G}'}$

convolution : double sum on $\boldsymbol{G}$ space ➔ $\mathcal{O}(N\,M^2_{PW})$!!

Instead:

$u_{m,\,\boldsymbol{k}+\boldsymbol{G}}$

↓ FFT^{-1} $\mathcal{O}(N\,M_{PW}\log M_{PW})$

$u_{m\,\boldsymbol{k}}(\boldsymbol{r})$ → $|u_{m\,\boldsymbol{k}}(\boldsymbol{r})|^2$

…. sum up on $\boldsymbol{k}$-points and states

…→ $n(\boldsymbol{r})$

↓ FFT

$n(\boldsymbol{G})$

Numerically calculate the energy functional DFT terms

- The DFT functional

$$E[n] = T_s[n] + \int d^3r\, V_{\text{ext}}(\vec{r})\, n(\vec{r}) + E_{\text{H}}[n] + E_{\text{xc}}[n]$$

- The (non-interacting) kinetic energy

$$T_s[n] = 2\sum_m \sum_{\vec{k}} w_{\vec{k}} f_{m,\vec{k}} \int d^3r\, \psi^*_{m,\vec{k}}(\vec{r}) \left[-\frac{\hbar^2\nabla^2}{2m}\psi_{m,\vec{k}}(\vec{r})\right]$$

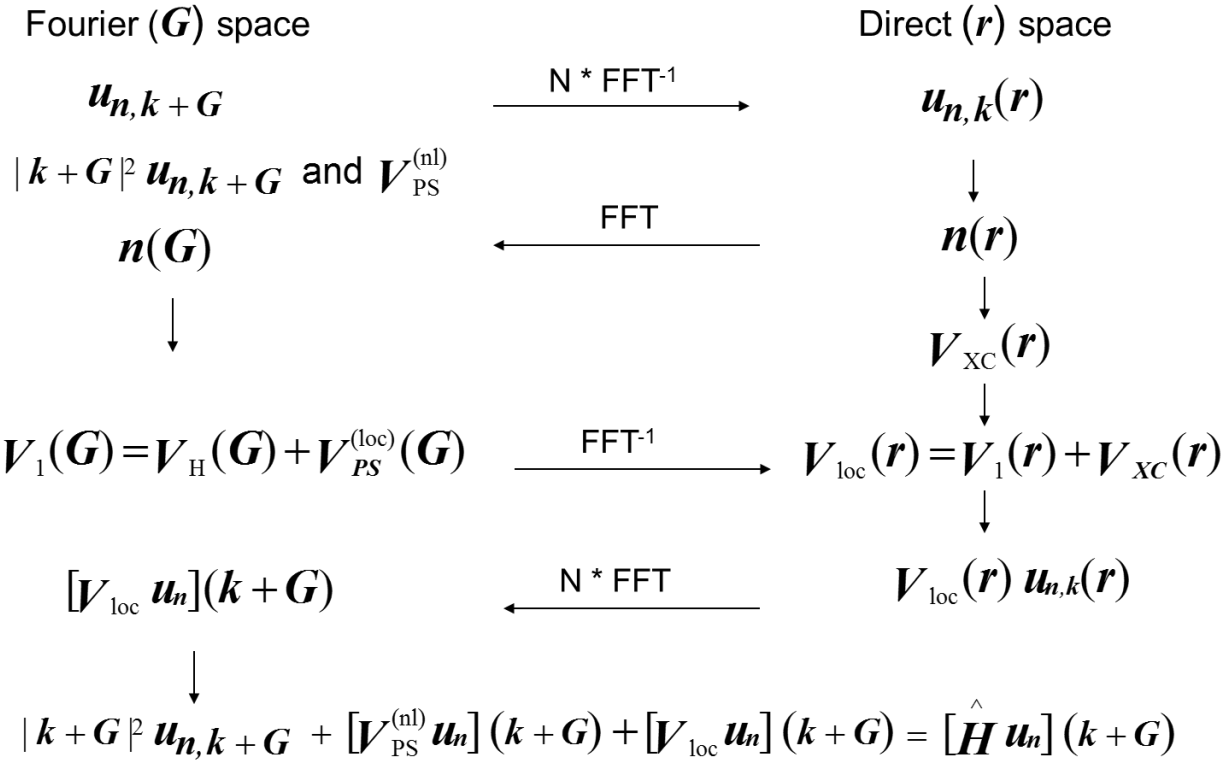
- The electrostatic (Hartree+pseudopotential) terms

$$E_{\text{H}}[n] = \frac{e^2}{2}\int d^3r \int d^3r'\, \frac{n(\vec{r})n(\vec{r'})}{|\vec{r}-\vec{r'}|} = 2\pi\Omega_0 \sum_{G\neq 0} \frac{|\tilde{n}(\vec{G})|^2}{G^2}$$

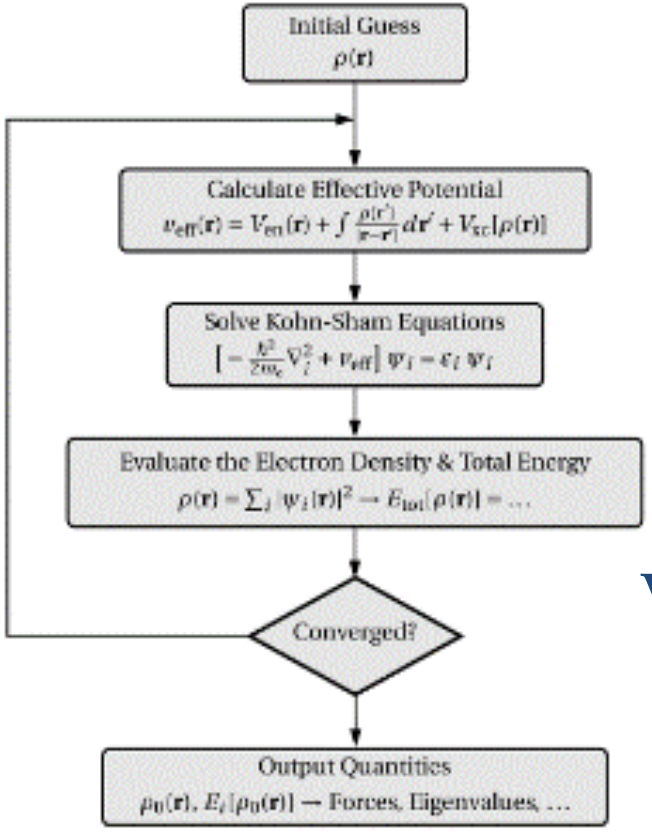
- The XC term

$$E_{\text{xc}}[n] \simeq E_{\text{xc}}^{\text{LDA}}[n] = \int d^3r\, n(\vec{r})\, e_{\text{xc}}^{\text{HEG}}(n(\vec{r}))$$

Numerical solutions: making us of direct and reciprocal spaces and Fast Fourier Transforms



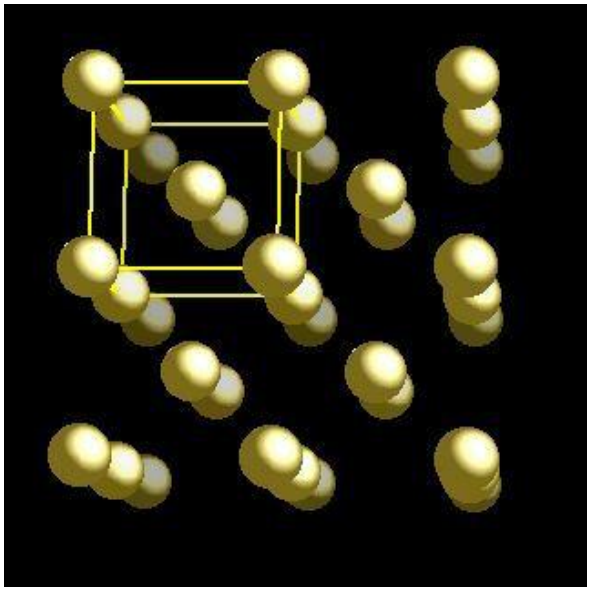
Numerical solutions: DFT flowchart (again)

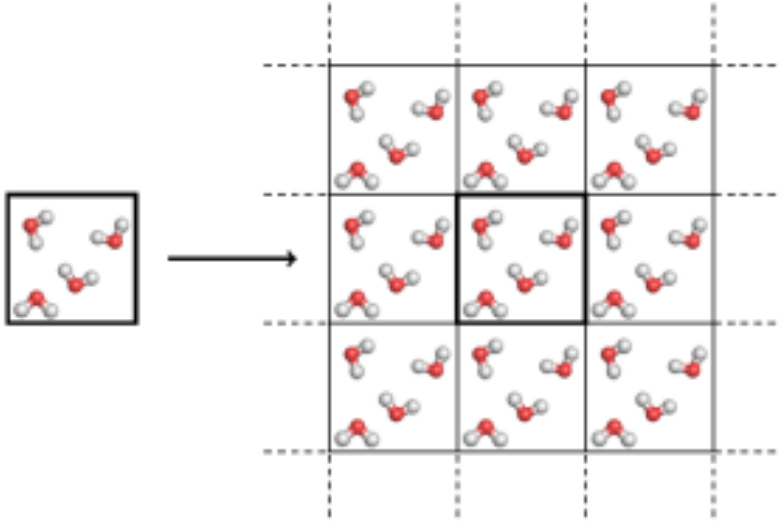


$$V_{new} = \beta V_{new} + (1 - \beta) V_{old}$$
$$\beta < 1; \beta \approx 0.7$$

A word on periodic boundary conditions

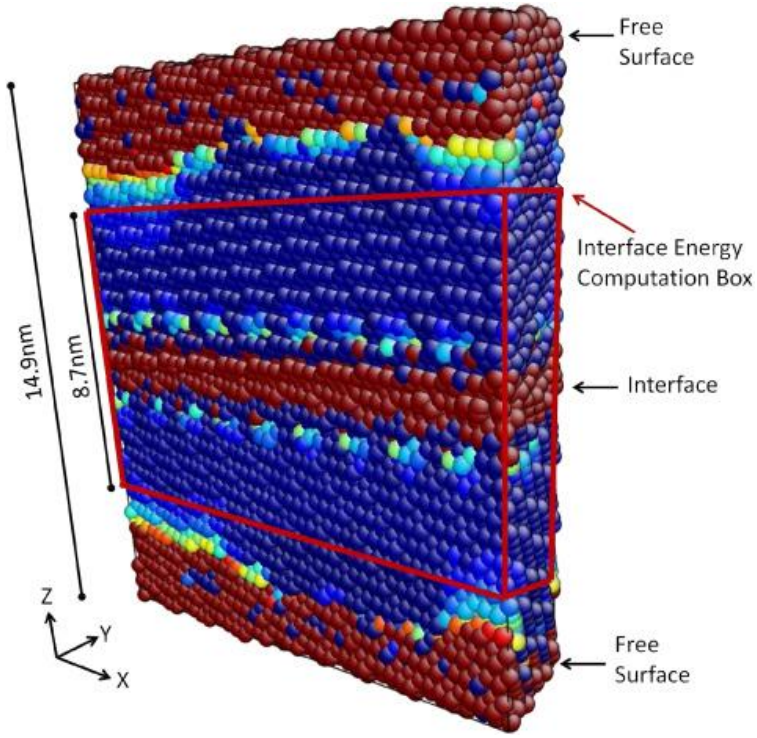
- DFT codes make large use of PBC, but their use in non 3D-bulk/non-crystalline systems is tricky





A word on periodic boundary conditions

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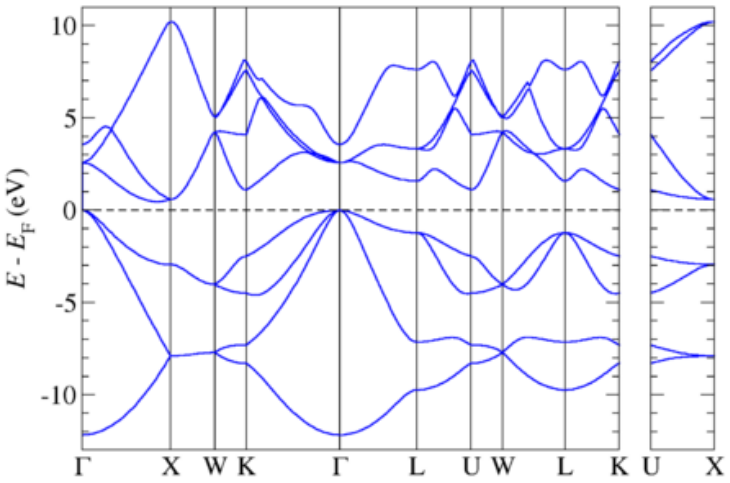


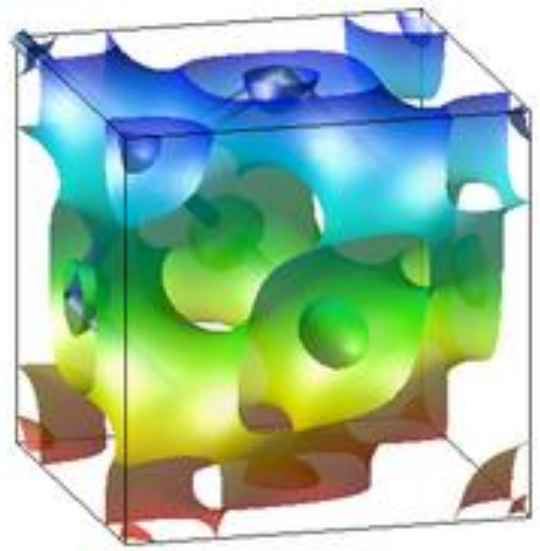
Plane-waves: take-home message

- Take advantage of crystal symmetry and periodicity
- Independent of atomic positions
- Same precision everywhere/useless accuracy in void space
- Formulae are easy to implement
- Choice between R and G spaces/Take advantage of fast Fourier transforms
- Many PW's (~10^2 – 10^3/atom): cutoff choice needed, easy to check/improve convergence
- Generally used in conjunction with pseudopotentials (no all-electron calculations just using PW's)

"Total energy" basic properties

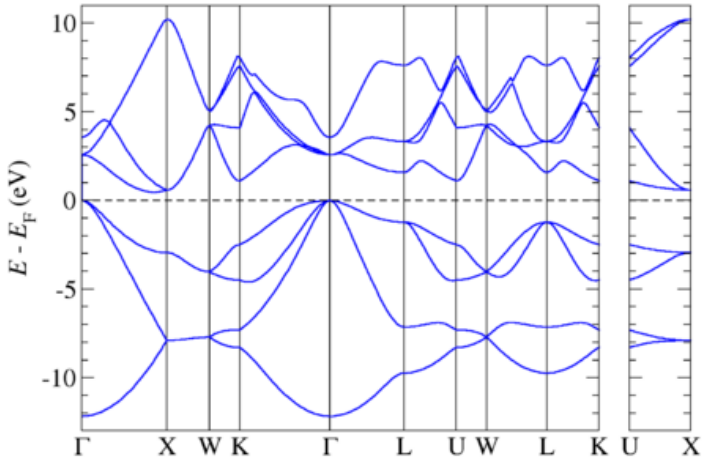
- **Electronic band structure and charge density map**

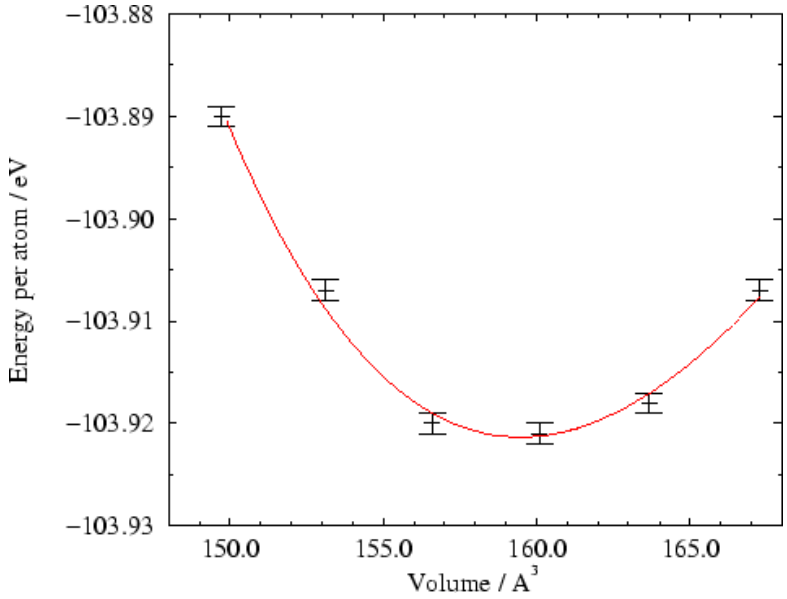




"Total energy" basic properties

- **Energy-volume equations of state**



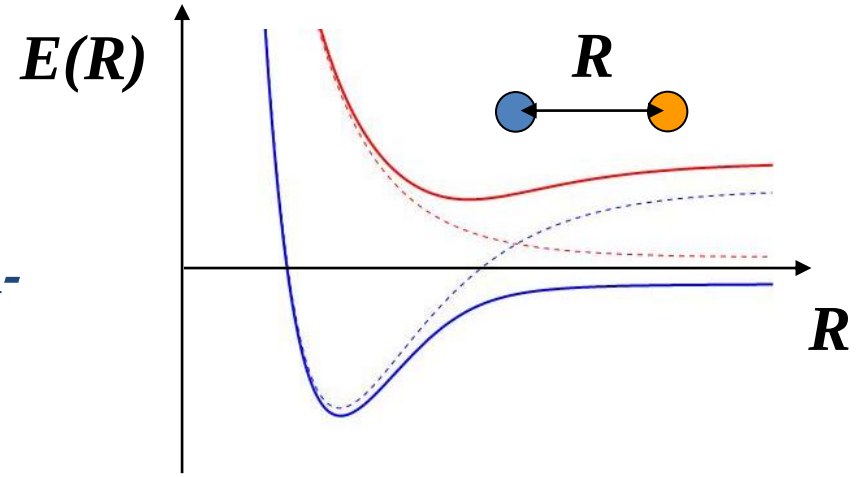


Derivatives of DFT energy: forces and pressure (stress)

- **Forces**

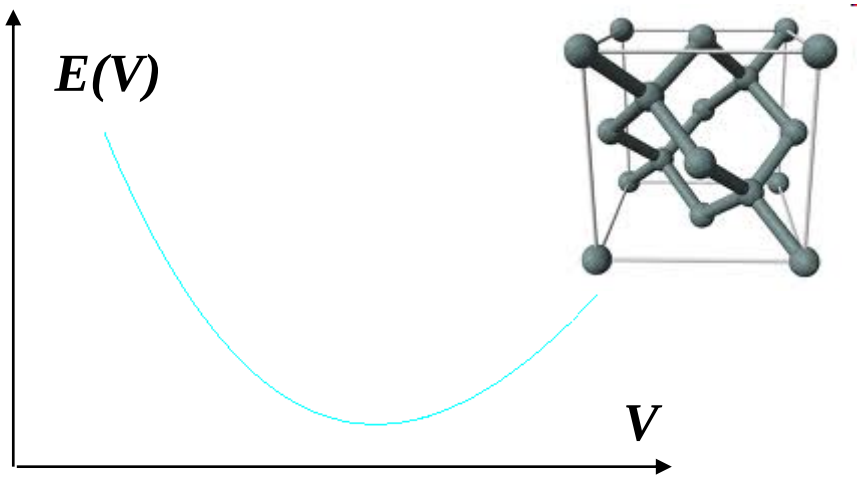
$$F = -\frac{\partial E}{\partial R}$$

Forces on the nuclei, within the Born-Oppenheimer approx. ➔ find the optimal bond length R_0



Derivatives of DFT energy: forces and pressure (stress)

- **Pressure**



$$P = -\frac{\partial E}{\partial V}$$

Pressure ➔ find the optimal volume V_0 of the unit cell

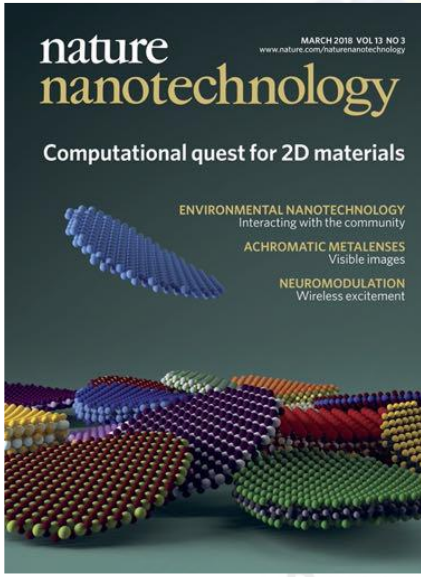
Derivatives of DFT energy: forces and pressure (stress)

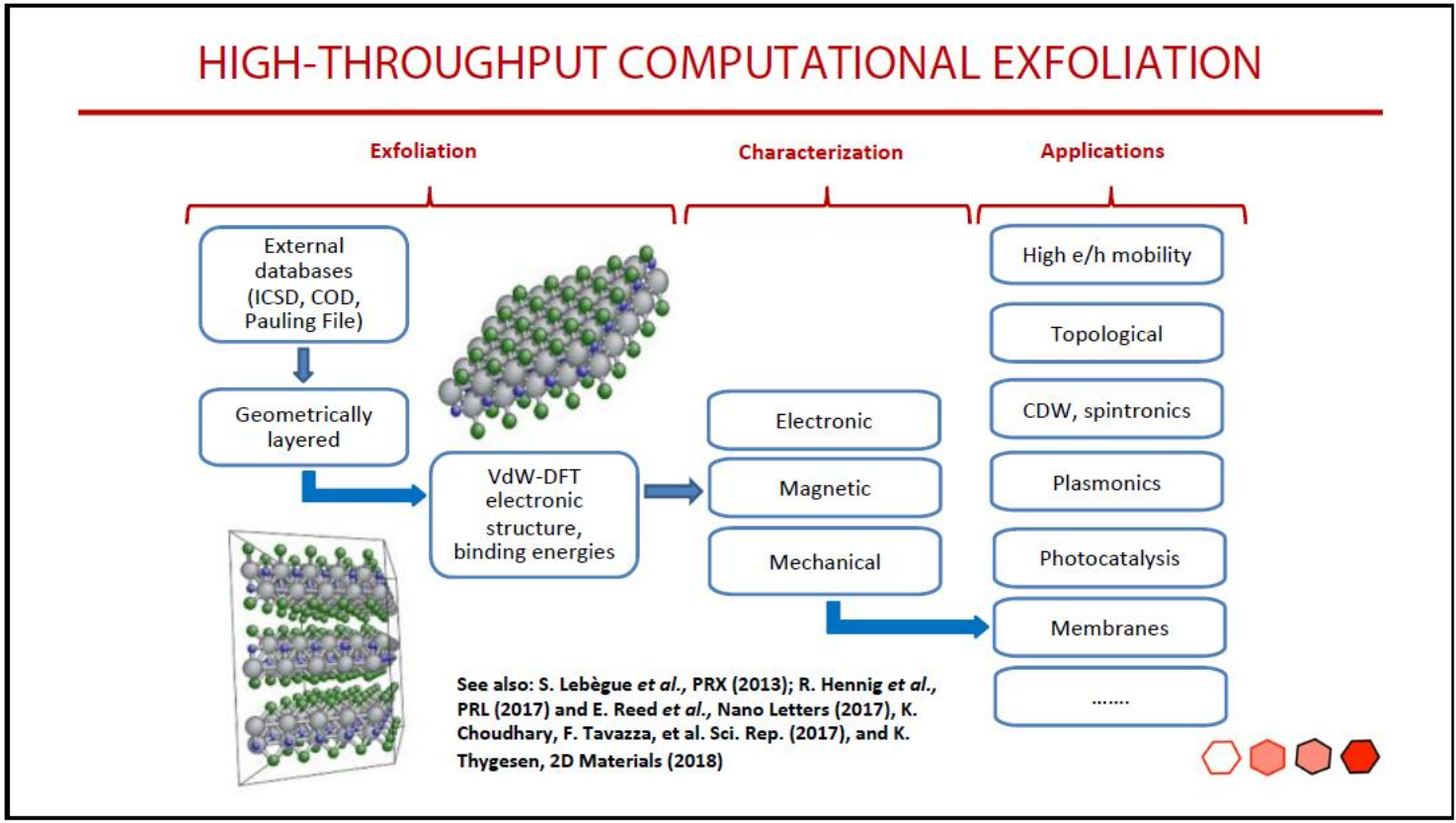
- **Hellmann-Feynman theorem**

$$\frac{\mathrm{d}E_\lambda}{\mathrm{d}\lambda} = \frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi_\lambda|\hat{H}_\lambda|\psi_\lambda\rangle = \left\langle\psi_\lambda\left|\frac{\mathrm{d}\hat{H}_\lambda}{\mathrm{d}\lambda}\right|\psi_\lambda\right\rangle$$

$$\frac{dE}{d\vec{R}_J} = \frac{\partial E}{\partial \vec{R}_J} + \int d^3r \frac{\delta E}{\delta n(\vec{r})}\frac{\partial n(\vec{r})}{\partial \vec{R}_J} = \frac{\partial E}{\partial \vec{R}_J}$$

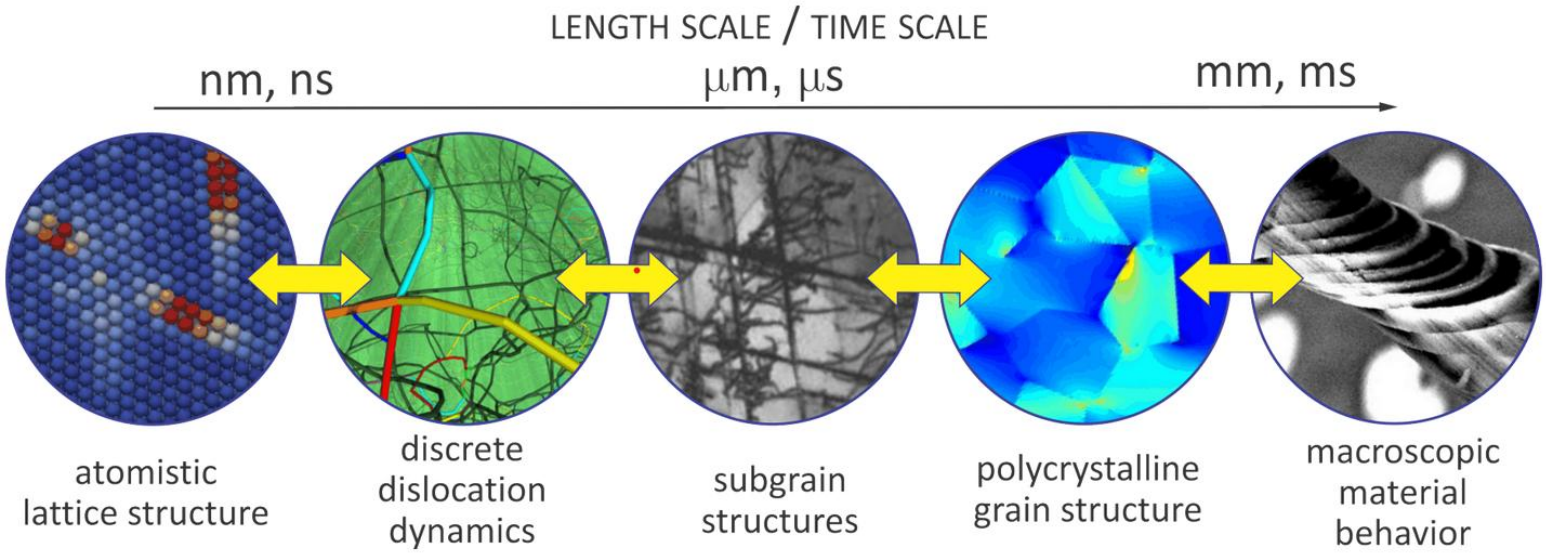
HPC for materials science

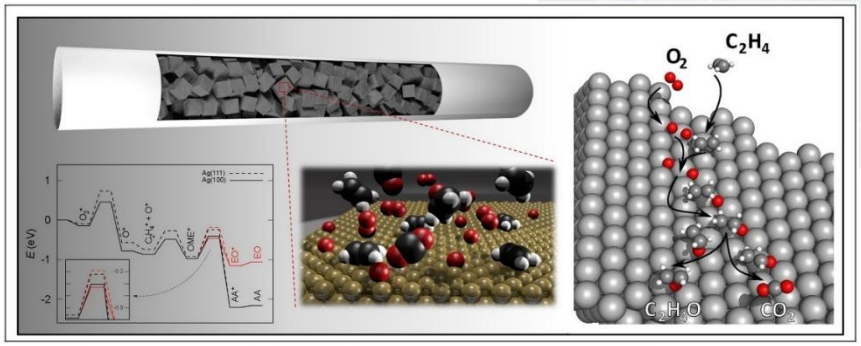


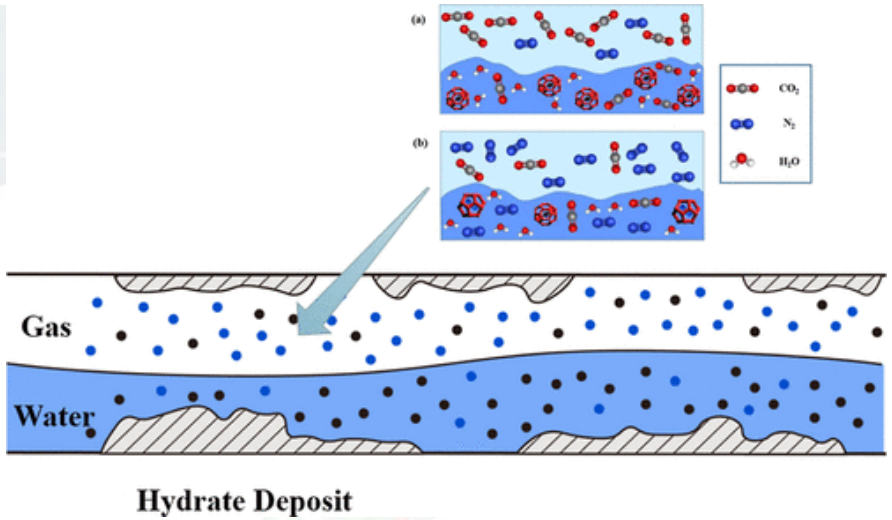


Courtesy of Nicola Marzari, EPFL

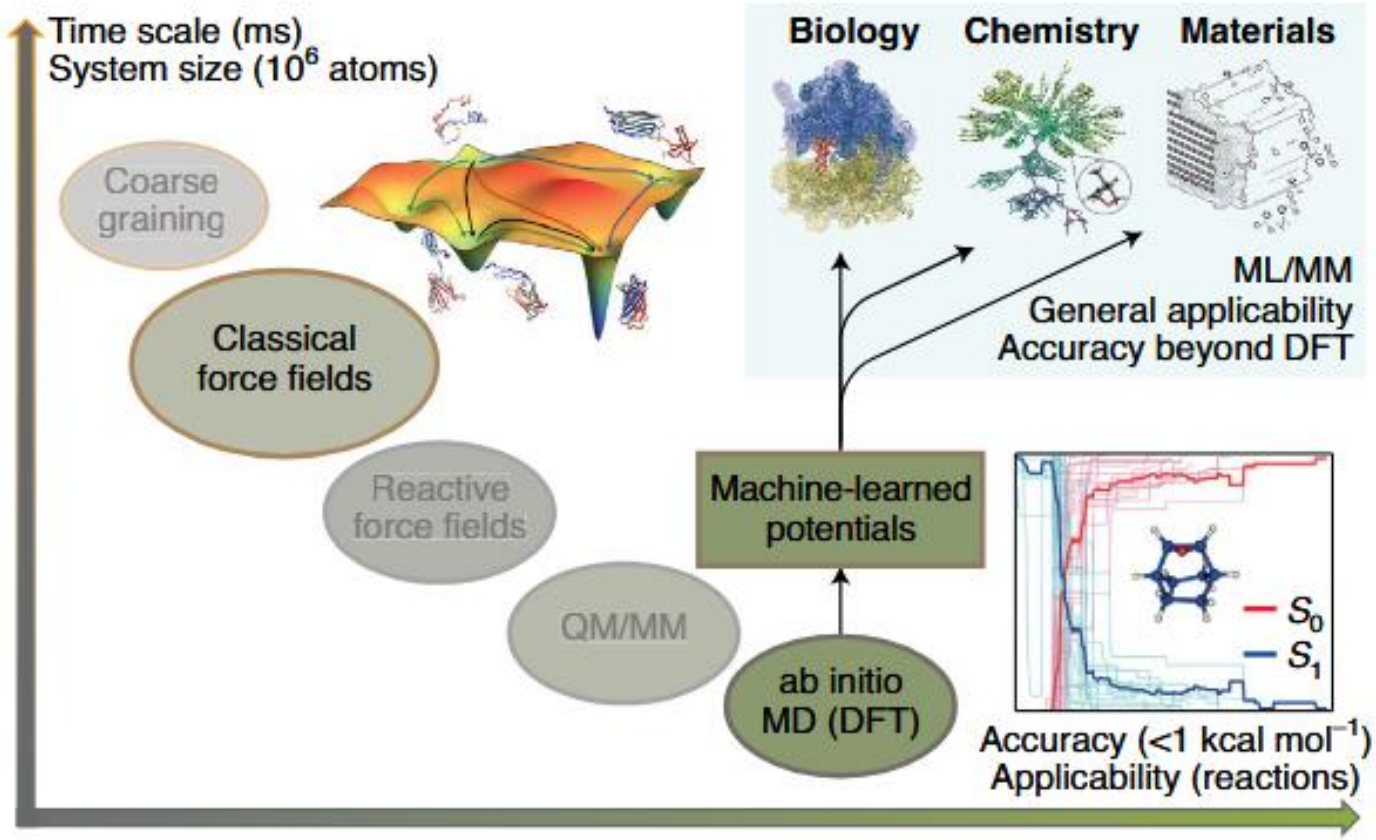
«Real» materials science







Length/time scales, methods



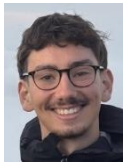
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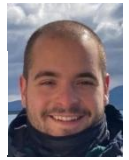
- **IA & Materials:**
  - Agustin Salcedo (Engineer PEPR DIADEM)
- **Materials for energy:**
  - Zacharie Waysenson (PhD PEPR Batteries)
  - Fortunata Panzera (PhD Università di Messina)
  - Marko Vukasinovic (PhD CNRS International)
  - Baptiste Barthe--Gold (PhD DGA)
- **CO_2 sequestration/heterogeneous catalysis:**
  - Stefano Ferrero (Postdoc DIM MaTerRE)
  - Gerard Pareras (Postdoc Marie-Curie/UE)
- **Hydrogen storage:**
  - Loan Renaud (PhD Sorbonne U)
  - Ameer Baloch (PhD SCAI Sorbonne U)











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DIADEM SUMMER SCHOOL 2026

Density Functional Theory: General Concepts

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DIADEM School – Grenoble, Aug 31st 2026

